

Generating Sketch-Based Synthetic Seismic Images With Generative Adversarial Networks

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Abstract—The characterization of the subsurface is paramount for the exploration and production life cycle and, more specifically, for the identification of potential hydrocarbon accumulations. In recent years, the oil and gas industry has increased its interest in applying machine learning to accelerate the seismic interpretation process, which is regarded as a time-consuming and human-centered task. Although machine learning has been successfully used in many applications ranging from stratigraphic segmentation to salt dome detection, a usual bottleneck is the need for a large amount of high-quality annotated data. To overcome this, data augmentation approaches are commonly used, and one of the most powerful is synthetic data generation. In addition, sketch-based synthetic seismic images can be used to support image retrieval applications to help oil companies leverage petabytes of seismic data sets. In this context, this letter investigates the generation of synthetic seismic images based on sketches using generative adversarial networks (GANs). To the best of our knowledge, this is the first work to propose such an approach. Experiments with five different sketch types in a public seismic data set indicate that realistic seismic images can be synthesized using rather simple sketches.

Index Terms—Artificial neural networks, geophysical image processing, image generation.

I. INTRODUCTION

SKETCH-BASED modeling (SBM) applications are quite common in geosciences and other domains. In 2012, Amorim *et al.* [1] proposed an SBM of seismic horizons with adaptive meshes and operators tailored for the geological domain. In the same year, Lidal *et al.* [2] proposed a novel sketch-based graphical approach which allows the visualization and examination of geological models' evolution. Lidal *et al.* [3] compare the use of many specialized modeling functions for 3-D sketch-based geological modeling with a single generic model. More recently, Hu *et al.* [4] explore the use of sketches in combination with uncertainty information to generate geological surface realizations through the diffusion process. As an example from another domain, but with a clear application to the oil and gas industry, Lei *et al.* [5] proposed the use of a generalized boundary detector [6] to extract boundaries from images and train a convolutional neural network (CNN) for a sketch-based image retrieval (SBIR) application using the Flickr15k data set.

In this letter, we propose the use of sketches for the generation of synthetic seismic images using Generative Adversarial

Networks (GANs). This letter greatly extends a previous investigation by Ferreira *et al.* [7] which considered only one edge-based sketch type. Sketch-based synthetic seismic images are a powerful tool to communicate what-if scenarios, support educational initiatives, retrieve images from databases and documents, and augment machine learning training data sets. Regarding the retrieval application, such a tool can be used to translate from hand-sketches to realistic seismic images that could then be used as input to traditional content-based image retrieval (CBIR) systems. This approach is more flexible than the aforementioned SBIR because it does not require a sketch-specific retrieval system.

As for the data augmentation application, it seems an interesting fit for the increasing interest in machine learning by the oil and gas industry. Many authors have investigated machine/deep learning methods to assist different tasks in subsurface characterization. Chevatarese *et al.* [8] present a deep convolutional network for stratigraphic segmentation. Zeng *et al.* [9] and Karchevskiy *et al.* [10] use a segmentation process to delineate salt domes. Halpert *et al.* [11], Oliveira *et al.* [12], and Lipari *et al.* [13] investigate the use of GANs to improve seismic image resolution, emulating a denser acquisition geometry. In addition, Lipari *et al.* [13] investigate the same methodology to obtain the deconvolved reflectivity image. Targeting a different application, Lu *et al.* [14] and Pochet *et al.* [15] investigate CNNs for fault detection. The main contribution of the former is to use GANs to increase the seismic image resolution before the classification task, while the latter relies on synthetic data to train the classification network.

Despite the promising results, a common issue permeates all these machine learning methods: the demand for a large amount of high-quality annotated data and, consequently, for time and human resources to produce them. Although Pochet *et al.* [15] rely on a library to create simple synthetic seismic data with faults, the creation of realistic seismic images with more complex structures such as channels, gas chimneys, and salt domes is only possible through the use of acoustic wave propagation models, which are computationally expensive and even impractical for some applications [16]. Moreover, they require deep knowledge of partial differential equations, forward and inverse models, and the generation of realistic velocity models. In this letter, we investigate an alternative method that relies on simple sketches and GANs to quickly generate realistic seismic images without the burden of dealing with physical simulators.

II. METHODOLOGY

In the past few years, GANs received a lot of attention in the computer vision community. They were introduced by

Manuscript received September 2, 2019; accepted October 1, 2019. Date of publication October 15, 2019; date of current version July 22, 2020. (Corresponding author: Rodrigo S. Ferreira.)

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Digital Object Identifier 10.1109/LGRS.2019.2945680

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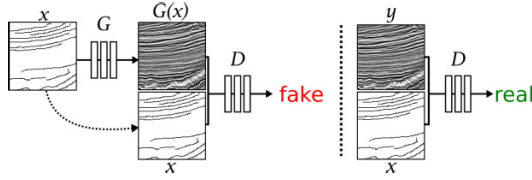


Fig. 1. In the adversarial schema investigated in this letter, the generator G receives a sketch image x and synthesizes a seismic image $G(x)$, while the discriminator D receives a pair of images and decides whether it is real (composed by x and the original image y) or fake (composed by x and $G(x)$).

Goodfellow *et al.* [17] in 2014 and several modifications and improvements were proposed since then. The basic idea of adversarial training involves two networks: a generator (G) that is responsible for synthesizing images (y) and a discriminator (D) that learns if a given image is real or synthesized, as depicted in Fig. 1. In this optimization schema, G is trained to generate images realistic enough to fool D , while D learns to discriminate between images synthesized by G and real images.

Considering a training set distribution $p_{\text{data}}(x)$, the trained generator learns a similar distribution $p_{\text{model}}(w)$ such that the discriminator is actually not able to detect if a given sample came from $p_{\text{data}}(x)$ or $p_{\text{model}}(w)$. The optimization formulation for finding the optimal mapping function G^* is classically defined as

$$G^* = \arg \min_G \max_D \mathcal{L}_{\text{GAN}}(G, D) \quad (1)$$

where $\mathcal{L}_{\text{GAN}}(G, D)$ is the GAN objective function defined by

$$\mathcal{L}_{\text{GAN}}(G, D) = E_{x \sim p_{\text{data}}(x)} [\log D(x)] + E_{z \sim p(z)} [\log(1 - D(G(z)))] \quad (2)$$

where E and $\log$ are the expectation and logarithmic operators, respectively, and z is an array of random values from a white noise distribution $p(z)$.

Conditional GANs (cGANs) were a natural and very successful extension of GANs [18]. Here, the generator learns to create images from a domain (y) using images from another domain (x), and the discriminator learns to identify if a given image from y is real or synthesized, given a pair of images from x and y . The loss function for cGANs is presented

$$\mathcal{L}_{\text{cGAN}}(G, D) = E_{x, y \sim p_{\text{data}}(x, y)} [\log D(x, y)] + E_{x \sim p(x), z \sim p(z)} [\log(1 - D(x, G(x, z)))] \quad (3)$$

The solution of (3) is achieved using an alternate training schema, as in classic GANs: the generator training uses the loss obtained using the last trained discriminator, while the discriminator computes losses using images synthesized by the last trained generator. The idea is that at each cycle, the discriminator receives more realistic synthesized images which improve its discriminative power, and the generator is optimized to provide more realistic images. At convergence, one expects a trained generator to produce very realistic images.

This letter uses the Pix2Pix cGAN architecture, which was introduced by Isola *et al.* [18]. This architecture implements a generator with an encoder–decoder architecture [19] and a “PatchGAN” [18] discriminator that uses patches to incorporate local information to the loss function.

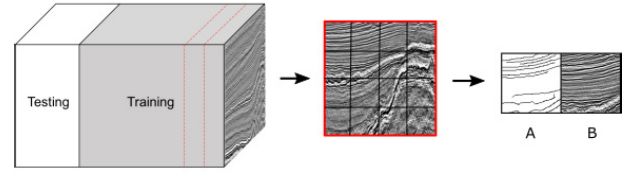


Fig. 2. Training and testing schema. The seismic cube was divided into two parts, one for training (80%) and one for test (20%). Then, crosslines were uniformly selected from each part to undergo a process of edge detection, for the edge-based sketches, and one that assigned different colors to the interpreted rock layers (strata) for the layer-based ones. Finally, the images were divided in 240×240 tiles, deriving test/training samples consisting of a pair of images: A—sketch image tile and B—target image tile with the corresponding seismic data.

In our experiments, we artificially generate sketch-like images and train the Pix2Pix network to synthesize the original image based on the sketch. The generator receives sketch images and tries to synthesize realistic seismic images, while the discriminator is trained to identify whether a given seismic image was artificially generated, as shown in Fig. 1.

III. EXPERIMENTS

A. Training and Test Data

In our experiments, we used a public seismic data set called Netherlands Offshore F3 Block [20]. The survey was acquired in 1987 in the North Sea at the coordinates N 54°52′0.8″/E 4°48′47.07″. The cube covers an area of 384 km² with 651 inlines and 951 crosslines. The seismic data has a range of 1848 ms, a sampling rate of 4 ms, and a bin size of 25 m. To build our training and test sets, we split the cube into two parts that were sampled uniformly. For the training set, we extracted 196 crosslines (~80%) and 50 crosslines were extracted for the test set (~20%), as shown in Fig. 2. Inlines could also be used, without any changes to the methodology.

After this step, we artificially generated five different types of sketches for the training and test sets: black edges (sketch A), layers (sketch B), colorful edges (sketch C), layers and black edges (sketch D), and layers and colorful edges (sketch E) [see Fig. 5(b)–(f) and (n)–(r)]. For all the sketch types that include edges, we applied the Canny edge detector algorithm to extract the edges automatically. We used scikit-image’s implementation [21] for which we provided the parameters $\text{sigma} = 3$, $\text{low threshold} = 0.25$, $\text{high threshold} = 0.75$, and $\text{use quantiles} = \text{true}$. The parameter $\text{use quantiles} = \text{true}$ makes the algorithm treat the parameters low and high thresholds as quantiles of the edge magnitude. These parameters were chosen so as to produce the most discriminating edges without overburdening the user with a too detailed sketch requirement.

As for the sketch types that include layers or colorful edges, we used an interpretation performed by two geoscientists as input. This interpretation produced nine horizons (see Fig. 3). It is noteworthy that the interpreted horizons may not correspond to the top of formations or stratal interfaces since only pattern configurations were taken into consideration, focusing on the separation of different seismic facies intervals. One fault was also interpreted just to assist the horizons interpretation as described in [22].

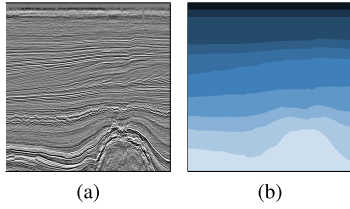


Fig. 3. Original seismic subset and all interpreted rock layers. (a) Seismic. (b) Rock layers.

Finally, we extracted 240×240 tiles from the training and test sets and reshaped them to 256×256 pixels allowing the use of deeper networks. Similarly, 34 000 training samples were created per sketch type, and 750 for the test. The training/test samples consist of a pair of images: (A) a sketch image tile and (B) a target image tile with the corresponding seismic data.

B. Training Scheme

Five networks were trained, one for each sketch type, using the same configuration. The networks were trained for 25 epochs, considering 34 000 samples taken at random. Adam optimizer was used with an initial learning rate of 0.0002 and a momentum of 0.5. The parameters for the generator loss function were set to GAN weight = 1 and L1 weight = 100. More information on the loss functions can be found in [18].

C. Test Results

To evaluate the results of the networks trained for each sketch type, we performed a quantitative and qualitative evaluation. In the quantitative evaluation, we computed the mean squared error (mse), structural similarity (SSIM), and Euclidean distance based on the Local Binary Pattern (LBP) texture attribute between the original seismic image and the one synthesized by the networks. The qualitative evaluation was performed by a geoscientist who examined the visual consistency of the generated seismic data.

1) *Quantitative Analysis:* In this analysis, we took the 750 test seismic images synthesized by each network and compared with the original seismic images. While mse defines a pixel-wise distance between two images, SSIM measures the similarity between two images by incorporating some texture information such as variances and co-variances. Since SSIM values are defined in the range $[-1, 1]$, to represent it as a distance, we used the common transformation $DSSIM(x, y) = (1 - SSIM(x, y))/2$. Finally, we also computed a more robust texture attribute – LBP – which has been successfully applied in medical images and geosciences [23], [24]. LBP was computed using four neighbors, distance 1, and tile size of 64 pixels. To compute the distance between two images, we used the Euclidean distance between the LBP vectors. The results are shown in Table I.

Overall, the networks trained on sketch types D and E were the ones that produced the best results. For DSSIM and LBP, sketch type D produced the best result with median distances of 0.39 and 0.17, respectively. For mse, sketch type E produced the lowest distance with a median value of 4712.1. The distance quartiles are reported to give a more detailed view of how each network performed given that each layer being

TABLE I
MSE, SSIM, AND LBP DISSIMILARITY VALUES FOR ALL SKETCH TYPES

	sketch A	sketch B	sketch C	sketch D	sketch E
MSE					
min	4395.2	4694.8	4131.9	3685.4	3720.4
q1	5566.5	5372.3	5019.3	4519.3	4495.6
median	6073.1	5628.2	5387.6	4803.0	4712.1
q3	6341.3	5987.9	5686.7	5024.4	4914.1
max	7517.4	6807.6	6532.6	5968.0	5931.6
DSSIM					
min	0.3849	0.4137	0.3781	0.3381	0.3416
q1	0.4398	0.4458	0.4144	0.3799	0.3864
median	0.4675	0.4565	0.4317	0.3978	0.4029
q3	0.5000	0.4652	0.4541	0.4159	0.4154
max	0.5093	0.4904	0.4941	0.4528	0.4635
LBP					
min	0.1453	0.1374	0.1401	0.1222	0.1311
q1	0.1832	0.1872	0.1825	0.1583	0.1623
median	0.2011	0.2074	0.1970	0.1714	0.1780
q3	0.2276	0.2463	0.2175	0.1865	0.1932
max	0.3875	0.3410	0.3145	0.2428	0.2510

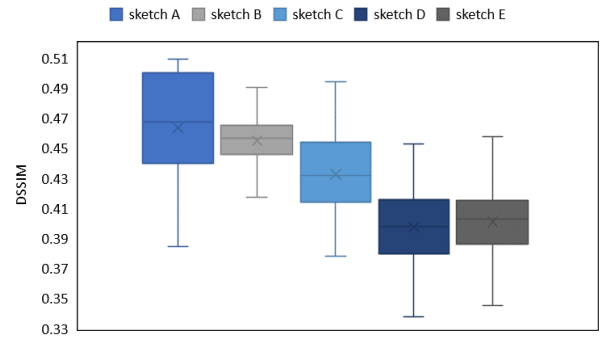


Fig. 4. Box plot of DSSIM values for all sketch types.

synthesized pose different levels of difficulty. For example, we show the box plot for the results with DSSIM in Fig. 4.

The plot shows an increasing performance from sketch types A to E, with sketch type D being a little better than E. The higher distance range for sketch type A may be explained by the difficulty of the network trained with this sketch type had to synthesize salt domes. This is expected since the network is trying to translate edges with the same color and size to different textures and amplitudes. Nevertheless, this sketch type produced convincing results for regions with less complexity.

Sketch type B produced the second worst result. Although introducing layer information improved a little the performance, it seems that the lack of information about the edges made it more difficult for the network to generate convincing textures. However, the low distance range indicates that the performance was more homogeneous throughout the different layers. By using colorful edges, sketch type C improved the performance showing that the combination of edges and layers presents a more powerful sketch representation. This combination was further investigated in the last two sketch types, D and E, which produced the best results. These results showed that using the layer information in combination with edges was the best configuration among the sketch types considered, reducing the error to 15% in relation to simple edges (sketch type A).

2) *Qualitative Analysis:* After analyzing the performance of the proposed sketch types in terms of image similarity, we conducted a visual analysis of the quality of the synthesized

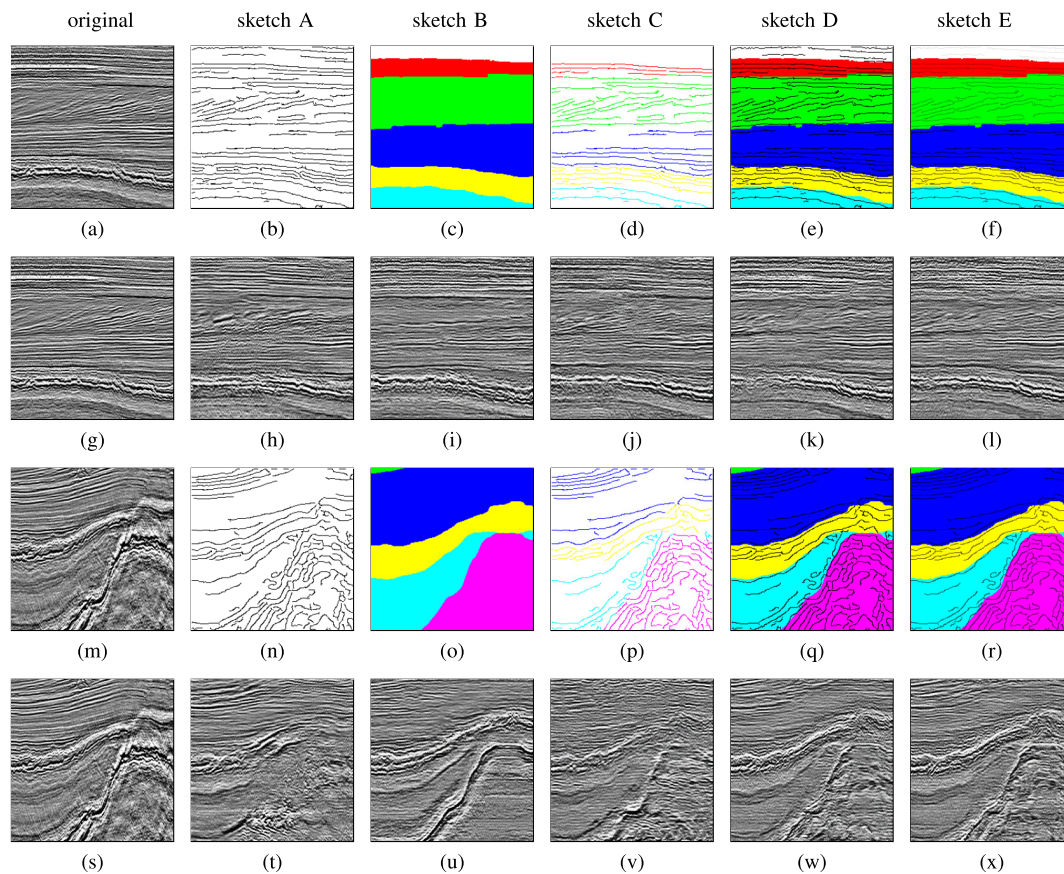


Fig. 5. (a) Test subset 1 from crossline 1116 and (b)–(f) respective input sketches and (h)–(l) synthetic seismic images. (m) Test subset 2 from crossline 1196 and (n)–(r) respective input sketches and (t)–(x) synthetic seismic images. (g) Test subset 1. (s) Test subset 2.

seismic images (see Fig. 5). The first two rows show the input sketches (b)–(f) and synthesized images (h)–(l) for test subset 1 from crossline 1116 (a). In this subset, we can see six layers comprising different facies with parallel and subparallel reflectors with varying amplitudes. The green and yellow layers represent the most different characteristics with a prograding sigmoidal configuration and polygonal faults, respectively. The lithologies range from sandstones and claystones to argillaceous deposits [25].

We can observe that for these layers, all sketch types produced satisfactory results. Note that sketch type B [see Fig. 5(i)], for not considering edges, was not able to properly generate the sigmoidal configuration for this seismic subset. It also represents layers' internal facies with generic texture/reflectors which seem to be a global representation of the layer and do not reproduce local characteristics. On the other hand, the other sketch types present quite similar results, being able to properly synthesize the target seismic image. The network trained with sketch type E was the one that produced the most realistic output [see Fig. 5(l)].

These results indicate that for parallel reflectors with low amplitude variation, the simplest sketch types like A and B should be used, since they produce as good results as the more complex ones and are easier to make. This was not the case for test subset 2 from crossline 1196 [see Fig. 5(m)]. In this subset, we can observe four layers. Three of them were already presented in the previous analysis, but in this subset they appear deformed by the underlying salt dome. This salt

dome represents the bottommost interpreted layer. It limits the top of an evaporite-rich succession whose top reflectors feature stratified seismic facies while the lower ones present homogeneous seismic facies, where there are no apparent reflections [25].

We can see that because of their deformation, the top three layers that presented equivalent results for all sketch types in test subset 1, now are unsatisfactorily synthesized in subset 2 for sketch types A and C [see Fig. 5(t) and (v)]. This was mainly due to the fact that sketch type A uses the same color and size to represent dipping reflectors with different amplitudes. For example, the ones in the sigmoidal progradation and the ones in the deformed layers above the salt dome. The result is slightly improved with sketch type C; nevertheless, it still presents vanishing reflectors. The other sketch types (B, D, and E) produced similar results to the ones obtained with test subset 1 for the top three layers.

A similar conclusion is obtained when we analyze the salt layer. Sketch types A and C present vanishing or poorly delineated salt domes [see Fig. 5(t) and (v)]. Sketch type B presents a better synthetic image [see Fig. 5(s)]; however, as pointed out in the quantitative analysis, the lack of edges seem to have a high impact on its ability to generate realistic textures. This is clear if we compare the internal texture of the salt dome generated by sketch type B with the one in image subset 2 [see Fig. 5(u)]. Similar to the results obtained in the quantitative analysis, sketch types D and E produced the best synthetic salt domes, being able to properly delineate their

boundaries along with convincing internal textures. Again, the network trained with sketch type E was the one that produced the most realistic output [see Fig. 5(x)].

The results obtained for test subset 2 suggest that, for more complex environments, sketch D or E should be used. It is noteworthy that although the quantitative analysis indicates sketch type D as the one with the best overall performance (see Table I), closely followed by sketch type E, the performed qualitative analysis indicates the opposite, with sketch type E producing the most realistic seismic images. Most likely, the crosslines selected to be discussed in this letter may not represent the general performance of the networks, showing a specific case in which sketch type E performed better. Another explanation is that the performance difference between the two is not significant and both produce equivalent results on average. Finally, the reason may be related to the quality metrics used. While mse agrees with the qualitative analysis, SSIM and LBP diverge. This may be due to the textural characteristic of the last two, which consider the spatial distribution of pixel intensities instead of single values. This makes SSIM and LBP miss edges/reflectors differences, which represent most of the differences perceived in our qualitative analysis.

IV. CONCLUSION

In this letter, we argued that sketch-based synthetic seismic images may be an interesting approach for generating realistic seismic data that can be used to communicate ideas, search for similar structures, or support the creation of training data sets for supervised machine learning algorithms. We evaluated five different sketch types and showed that it was possible to obtain realistic seismic images with rather simple sketches. The combination of background colors—representing rock layers—and colorful edges was the sketch type that produced the best results considering the quantitative and qualitative analysis conducted. As future works, the authors would like to investigate the performance of the proposed methodology in different applications such as image retrieval and classification.

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